

ANMOL PRABHAKAR

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EDUCATION

University of Illinois, Urbana-Champaign | 4.0 GPA

B.S. in Bioengineering, Minor in Electrical Engineering | James Scholar Honors

Champaign, IL

Expected May 2028

SKILLS

Technical: Onshape, SolidWorks, Fusion 360, Python (Pandas/NumPy), FDM 3D Printing, DFM/DFA, Rapid Prototyping

Fabrication: Rotary Grinder, Vacuum Bagging, Carbon Fiber Layups, Bandsaw, Thermoforming, Circuit Assembly, Soldering

Languages: Hindi (Native), Spanish (Professional Working Proficiency)

EXPERIENCE

Attune Neurofeedback EEG Hardware Team

Sep 2025 – Present

Hardware Engineer

Champaign, IL

- Engineered modular cartridge enclosures in Onshape using **interference fits and detents** to achieve tool-less component retention.
- Iterated **7+ 3D-printed prototypes** with internal channels, reducing package volume by 15% and eliminating wire pinch points.
- Designed an adjustable **compliant strap assembly** to maintain uniform clamping pressure and scalp contact across head sizes.
- Evaluated **analog front-end schematics** and interconnects to isolate microvolt-level neural biosignals from external circuit noise.

Beverly Orthopedics

Jul 2024

Clinical Engineering Intern

Montebello, CA

- Fabricated **13 composite orthotic braces** utilizing vacuum bagging and selective hardware support to maximize structural rigidity.
- Post-processed composite blanks using a **rotary grinder** to refine shape directly within patient shoe boundary constraints.
- Operated **optical surface scanners** across 20+ patients to digitize complex plantar surface morphology for custom diabetic insoles.
- Facilitated clinical consultations by providing **real-time Spanish translation** for 20+ patients during clinical consultations.

Credence Prosthetics

Jun 2024

Prosthetics Intern

Gurugram, India

- Assembled **8 custom KAFOs and braces** by mechanically aligning bilateral side-rails directly along sagittal anatomical knee axes.
- Calibrated **drop-lock hinges to 0° extension**, ensuring structural joint stability under 500+ N cyclic stance loads without binding.
- Formed **positive plaster limb molds** and thermoformed rigid polycarbonate shells, relieving pressure points via precision grinding.
- Conducted **visual gait evaluations** over 10-meter walk trials to tune device joint alignment and mitigate user hyperextension.

PROJECTS

Smart Lighting Controller Hub | Embedded Systems, Circuit Design, Firmware

Sep 2026

- Engineered an **ESP8266 controller hub** combining multi-protocol IR optical transmission and 5-button RF remote emulation.
- Interfaced GPIOs with an **LM3046 NPN transistor array** using 1 kΩ series and 10 kΩ pull-down resistors to eliminate latching.
- Powered the modified RF remote PCB directly from the **3.3V power rail** over a unified common ground bus running serial macros.

7-DOF Tendon-Driven Bionic Hand | CAD, Embedded Systems, Kinematics

Jun 2026

- Designed a **7-DOF hand in CAD** pairing braided Dyneema flexor tendons with antagonistic elastic return cords.
- Engineered an articulated **3-DOF thumb mechanism** incorporating micro servos for palmar sweep, pivot, and distal flexion.
- Programmed a **TI MSP432 and Python GUI** to coordinate 7 synchronized PWM channels, executing preset grips and animations.

Autonomous Sand-Leveling Robot (HackIllinois) | Mechanical Design, Kinematics, Robotics

Feb 2026

- Secured **1st place in John Deere Mechathon track**, achieving the flattest overall sand surface autonomously across all divisions.
- Engineered a **parallelogram 4-bar linkage** in 36 hours, holding a parallel rake blading angle over a 75 mm vertical height stroke.
- Formulated link lengths to step leveling passes through aggregate at **3 cut depths** to eliminate dynamic motor current spikes.

LEADERSHIP & ACTIVITIES

Biomedical Engineering Society (BMES)

Sep 2025 – Present

Workshop Committee Member

Champaign, IL

- Planned and led **technical workshops for 30+ students** covering parametric CAD modeling, functional 3D printing, and DFM.
- Collaborated with committee members to develop **interactive lab modules** and coordinate build equipment for student attendees.